C - Amidakuji

分值：600 分

Problem Statement

> Prepare as few types of ladder lotteries as possible and combine them so that all permutations can be produced.
>
> 准备尽可能少的 ladder lottery 类型，将它们组合后可以生成所有排列。

This is an **interactive problem** (in which your program and the judge program interact via input and output).

本题是**交互题**（你的程序需要与评测程序通过输入输出进行交互）。

You are given a positive integer N.

给定一个正整数 N。

Choose any positive integer m and m permutations of $(1, 2, \dots, N)$, and output them. Let the i-th permutation be $P_i = (P_{i,1}, P_{i,2}, \dots, P_{i,N})$.

选择任意正整数 m 和 m 个 $(1, 2, \dots, N)$ 的排列，然后输出这些排列。记第 i 个排列为 $P_i = (P_{i,1}, P_{i,2}, \dots, P_{i,N})$。

Then, a permutation $Q = (Q_1, Q_2, \dots, Q_N)$ of $(1, 2, \dots, N)$ is given. Output a sequence of positive integers $A = (A_1, A_2, \dots, A_k)$ satisfying all of the following conditions.

接下来你会得到一个 $(1, 2, \dots, N)$ 的排列 $Q = (Q_1, Q_2, \dots, Q_N)$。输出一个满足以下所有条件的正整数序列 $A = (A_1, A_2, \dots, A_k)$。

- $0 \le k \le 2N^2$
- All elements of A are between 1 and m, inclusive.
- Starting from the sequence $R = (1, 2, \dots, N)$, performing the following operation for $i = 1, 2, \dots, k$ in order results in $R = Q$.
- $0 \le k \le 2N^2$

- A 的所有元素都在 1 到 m 之间（含两端）。
- 初始为序列 $R = (1, 2, \dots, N)$，按顺序对每个 $i = 1, 2, \dots, k$ 执行以下操作后，结果等于 $R = Q$。

- Let $p = P_{A_i}$. Simultaneously replace R_j with R_{p_j} for each $j = 1, 2, \dots, N$.
- 令 $p = P_{A_i}$。对每个 $j = 1, 2, \dots, N$，同时将 R_j 替换为 R_{p_j}。

Let $m_{\min}$ be the minimum value of m for which the above problem can be solved regardless of Q. Your output m must equal $m_{\min}$.

记 $m_{\min}$ 为无论 Q 是什么，上述问题都有解的 m 的最小值。你输出的 m 必须等于 $m_{\min}$。

More precisely

更准确地说

For a sequence of permutations $P = (P_1, P_2, \dots, P_m)$ of $(1, 2, \dots, N)$, we call P a **good sequence of permutations** if the following condition is satisfied:

对于由 $(1, 2, \dots, N)$ 的排列组成的序列 $P = (P_1, P_2, \dots, P_m)$，如果满足以下条件，我们就称 P 为**好排列序列**：

- For any permutation $Q = (Q_1, Q_2, \dots, Q_N)$ of $(1, 2, \dots, N)$, there exists a sequence of positive integers $A = (A_1, A_2, \dots, A_k)$ satisfying all of the following conditions.
  - $0 \le k \le 2N^2$
  - All elements of A are between 1 and m, inclusive.
  - Starting from the sequence $R = (1, 2, \dots, N)$, performing the following operation for $i = 1, 2, \dots, k$ in order results in $R = Q$.
    - Let $p = P_{A_i}$. Simultaneously replace R_j with R_{p_j} for each $j = 1, 2, \dots, N$.
- 对于任意一个 $(1, 2, \dots, N)$ 的排列 $Q = (Q_1, Q_2, \dots, Q_N)$，都存在一个正整数序列 $A = (A_1, A_2, \dots, A_k)$ 满足以下所有条件：
  - $0 \le k \le 2N^2$
  - A 的所有元素都在 1 到 m 之间（含两端）。
  - 初始为序列 $R = (1, 2, \dots, N)$，按顺序对每个 $i = 1, 2, \dots, k$ 执行以下操作后，结果等于 $R = Q$。
    - 令 $p = P_{A_i}$。对每个 $j = 1, 2, \dots, N$，同时将 R_j 替换为 R_{p_j}。

It is guaranteed that at least one **good sequence of permutations** exists. Thus, the minimum value $m_{\min}$ of the length m of a **good sequence of permutations** is determined. Your output m

must equal $m_{\min}$.
Note that your output P does not need to be a **good sequence of permutations**. Your submission is judged correct if you can output an A satisfying the conditions for the given Q.

题目保证至少存在一个**好排列序列**，因此**好排列序列**长度 m 的最小值 $m_{\min}$ 是确定的。你输出的 m 必须等于 $m_{\min}$。 注意你输出的 P 不需要是**好排列序列**。只要你能对给定的 Q 输出满足条件的 A，你的提交就会被判定为正确。

Constraints

- $3 \leq N \leq 500$
- Q is a permutation of $(1, 2, \ldots, N)$.
- All input values are integers.
- $3 \leq N \leq 500$
- Q 是 $(1, 2, \ldots, N)$ 的一个排列。
- 所有输入值均为整数。

Input/Output

This is an **interactive problem** (in which your program and the judge program interact via input and output).

本题是**交互题**（你的程序需要与评测程序通过输入输出进行交互）。

First, the judge gives you a positive integer N in the following format:

首先，评测程序会按照以下格式给你一个正整数 N：

N

Then, output your chosen positive integer m and m permutations of $(1, 2, \ldots, N)$ in the following format over $m+1$ lines. Here, the j-th element of the i-th permutation is $P_{i,j}$. Be sure to output a newline at the end.

然后，按照以下格式，用 $m+1$ 行输出你选择的正整数 m 和 m 个 $(1, 2, \ldots, N)$ 的排列。这里第 i 个排列的第 j 个元素为 $P_{i,j}$。输出末尾必须换行。

m
$P_{1,1}\ P_{1,2}\ \ldots\ P_{1,N}$

$P_{2,1}\ P_{2,2}\ \ldots\ P_{2,N}$

$\vdots$

$P_{m,1}\ P_{m,2}\ \ldots\ P_{m,N}$

The following conditions must be satisfied:

必须满足以下条件：

- $m \leq 1000$
- $(P_{i,1}, P_{i,2}, \ldots, P_{i,N})$ is a permutation of $(1, 2, \ldots, N)$.
- $m \leq 1000$
- $(P_{i,1}, P_{i,2}, \ldots, P_{i,N})$ 是 $(1, 2, \ldots, N)$ 的一个排列。

If your output P does not satisfy the above conditions, the judge outputs `-1`. At that point, your submission has already been judged as incorrect. The judge program terminates at that point, so it is advisable for your program to terminate as well.

如果你的输出 P 不满足上述条件，评测程序会输出 `-1`。此时你的提交已经被判定为错误，评测程序会终止，因此建议你的程序也立即终止。

Then, the judge gives you a permutation $Q = (Q_1, Q_2, \ldots, Q_N)$ of $(1, 2, \ldots, N)$ in the following format:

接下来，评测程序会按照以下格式给你一个 $(1, 2, \ldots, N)$ 的排列 $Q = (Q_1, Q_2, \ldots, Q_N)$：

$Q_1\ Q_2\ \ldots\ Q_N$

Then, output a sequence of positive integers $A = (A_1, A_2, \ldots, A_k)$ in the following format. Be sure to output a newline at the end.

然后按照以下格式输出正整数序列 $A = (A_1, A_2, \ldots, A_k)$。输出末尾必须换行。

$k\ A_1\ A_2\ \ldots\ A_k$

Your output is judged as correct if and only if the following conditions are all satisfied:

当且仅当以下所有条件都满足时，你的输出会被判定为正确：

- $m = m_{\min}$

- $0 \leq k \leq 2N^2$
- All elements of A are between 1 and m, inclusive.
- Starting from the sequence $R = (1, 2, \ldots, N)$, performing the following operation for $i = 1, 2, \ldots, k$ in order results in $R = Q$.
- $m = m_{\min}$
- $0 \leq k \leq 2N^2$
- A 的所有元素都在 1 到 m 之间（含两端）。
- 初始为序列 $R = (1, 2, \ldots, N)$，按顺序对每个 $i = 1, 2, \ldots, k$ 执行以下操作后，结果等于 $R = Q$。

- Let $p = P_{A_i}$. Simultaneously replace R_j with R_{p_j} for each $j = 1, 2, \ldots, N$.
- 令 $p = P_{A_i}$。对每个 $j = 1, 2, \ldots, N$，同时将 R_j 替换为 R_{p_j}。

Notes

- **After each output, insert a newline and flush the standard output. Failure to do so may result in a judge verdict of TLE.**
- **If you receive `-1`, terminate your program immediately. If you do, the judge verdict will be WA, but if you do not, the judge verdict will be indeterminate.**
- Terminate your program immediately after outputting your answer as well. Otherwise, the judge verdict will be indeterminate.
- Do not output unnecessary newlines, as they will be considered malformatted.
- The judge for this problem is not adaptive. The judge program determines Q before the interaction begins.
- **每次输出后都要换行并刷新标准输出。否则可能导致评测结果为 TLE。**
- **如果收到 `-1`，立即终止你的程序。这么做的话评测结果会是 WA，否则评测结果是不确定的。**
- 输出答案后也请立即终止程序，否则评测结果是不确定的。
- 不要输出多余的换行，否则会被视为格式错误。
- 本题的评测是非适应性的，评测程序会在交互开始前就确定 Q。

Sample Input/Output

The following is a case with $N = 3, Q = (2, 3, 1)$.

以下是 $N = 3, Q = (2, 3, 1)$ 的一个样例。

Input	Output	Explanation
3		N is given.
	2 2 1 3 3 2 1	Choose $m = 2, P_1 = (2, 1, 3), P_2 = (3, 2, 1)$ and output them.
2 3 1		Since P satisfies the conditions, $Q = (2, 3, 1)$ is given.
	2 2 1	$A = (2, 1)$ satisfies the conditions. R becomes $(1, 2, 3) \to (3, 2, 1) \to (2, 3, 1)$, which matches Q. Also, $m_{\min} = 2$ for $N = 3$, and $m = m_{\min}$ is satisfied. Thus, this output is judged as correct. Be sure to output k as well, all on one line.

输入	输出	说明
3		给出 N。
	2 2 1 3 3 2 1	选择 $m = 2, P_1 = (2, 1, 3), P_2 = (3, 2, 1)$ 并输出。
2 3 1		因为 P 满足条件，所以给出 $Q = (2, 3, 1)$。
	2 2 1	$A = (2, 1)$ 满足条件。R 变为 $(1, 2, 3) \to (3, 2, 1) \to$

输入	输出	说明
		$(2,3,1)$，与 Q 一致。 同时对于 $N=3$，有 $m_{\min}=2$，满足 $m=m_{\min}$。 因此该输出被判定为正确。 注意也要将 k 完整输出在一行内。